Sudharshan Ravi

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Summary

- **My Profile:** Computational Researcher at the University of Lausanne (UNIL) | Ph.D. in Chemical and Bioengineering (Focus on Computational Systems Biology) at ETH Zürich | Swiss resident (valid C permit) | Indian citizen | 10+ years in Bioinformatics Research | 7+ years in Metabolic Network Analysis | 5+ years as Lead Researcher
- **My work:** Integrative high-dimensional multi-omics data analysis in understanding the biology of aging and age-related diseases | Novel workflow for the analysis and integration of omics data in genome-scale metabolic models | Inferential analysis on the impact of socioeconomic and other exogenous gradients on transcriptomic patterns and health outcomes

Experience

Premier Assistant *Lausanne, Switzerland*

Department of Fundamental Microbiology (DMF), University of Lausanne (UNIL) *Mar. 2026 – Present*

- Developing constraint-based computational models to characterize metabolic interactions and resource exchange networks in microbial communities, enabling mechanistic prediction of community dynamics.

Bioinformatics Lead *London, UK*

Myolytics *Jan. 2025 – Feb. 2026*

- Tailored machine learning models to longitudinal multimodal molecular and biosensor data for pre-symptomatic prediction of muscle injury risk, translating predictive insights into personalized management strategies for professional athletes.

Postdoctoral Research Associate *Zürich, Switzerland*

Social Genomics, Jacobs Center for Productive Youth Development, University of Zurich *Mar. 2021 – Jan. 2025*

- Characterized molecular consequences of socioeconomic inequalities across transcriptomic, immunological and metabolic axes using multi-omics and network-based approaches on a longitudinal cohort spanning adolescence to adulthood.
- Revealed disruption of upstream immune regulatory networks as a key mechanism connecting early-life socioeconomic disparities to altered metabolic function, accelerated aging and increased disease predisposition.

Graduate Research Assistant / Doctoral Candidate *Zürich, Switzerland*

Chemical and Biological Systems Engineering Laboratory, ETH Zürich *Jan. 2016 – Sep. 2020*

- Bioinformatic analysis of human aging: Extensive analysis of population-level transcriptomic data characterizing the biology of aging and its close relationship to metabolism.
- ΔFBA: Developed a novel pipeline integrating differential omics into genome-scale metabolic models using constraint-based modeling and mathematical optimization.
- Multi-omics analyses: Deciphered the chromatin – gene expression interplay in cell fate commitment of hematopoietic cells. Illuminated the mechanisms of constrained antitumor immune responses during IDO1 inhibition in ovarian cancer.

Research Assistant *Boston, USA*

Sengupta's Lab for Nanomedicine, Harvard – MIT Health Sciences and Technology *Jan. 2012 – Aug. 2012*

- Elucidated CD44v6–EGFR scaffold kinase interaction as a mechanism predisposing chemotherapy-tolerant breast cancer cells and a bivalent therapeutic target for overcoming drug resistance.

Education

Ph.D. in Chemical and Bioengineering — *Zürich, Switzerland*

ETH Zürich | Visiting researcher at University at Buffalo, USA *(Aug. 2018 – Jul. 2020)* — *Jan. 2016 – Sep. 2020*

- Thesis: Metabolic network analysis and its application in understanding the biology of aging – Prof. Dr. Rudiyanto Gunawan
- Research Specialization: Bioinformatics | Systems Biology | Aging & Age-Related Diseases | Metabolic Network Analysis | Constraint-Based Modeling | Integrative Multi-omics

M.Sc. in Chemical and Bioengineering — *Zürich, Switzerland*

ETH Zürich — *Sep. 2013 – Sep. 2015*

- Thesis: Reducing Complexity — Efficient Modeling for Biorefinery Concepts – Prof. Dr. Konrad Hungerbühler
- Research Specialization: Surrogate Modeling | Sustainable Engineering | Machine Learning | Bioprocessing Technologies | Process Engineering

B.Tech. in Industrial Biotechnology — *Thanjavur, India*

SASTRA University | Harvard – MIT Health Sciences and Technology — *Sep. 2008 – Oct. 2012*

- Thesis: An Induced CD44v6 – EGFR scaffold kinase interaction predisposes a subpopulation of chemotherapy-tolerant breast cancer cells and serves as a bivalent target of resistance – Prof. Dr. Aaron Goldman
- Research Specialization: Molecular Biology | Genetics | Cancer Biology | Drug Resistance | Nanomedicine

Selected Publications

[1] **Sudharshan Ravi**, Michael J. Shanahan, Brandt Levitt, *et al.* Socioeconomic inequalities in early adulthood disrupt the immune transcriptomic landscape via upstream regulators. **Scientific Reports**, 2024

[2] **Sudharshan Ravi** and Rudiyanto Gunawan. ΔFBA – Predicting metabolic flux alterations using genome-scale metabolic models and differential transcriptomic data. **PLOS Computational Biology**, 2021

[3] Emelyne Teo, **Sudharshan Ravi**, Diogo Barardo, *et al.* Metabolic stress is a primary pathogenic event in transgenic *Caenorhabditis elegans* expressing pan-neuronal human amyloid beta. **eLife**, 2019

See complete list of publications and conference presentations at **Google Scholar**.

Proficiencies

Computational	R | Bioconductor | Python | scikit-learn | pandas | MATLAB | COBRA Toolbox
Infrastructure	Bash | Git | Docker | High-Performance Computing
Networks	Cytoscape | STRING | iGraph | Gephi
Visualization	ggplot2 | Adobe Illustrator | Inkscape | PowerPoint | LaTeX
Laboratory	Cell culture | IHC | Western Blotting | Flow cytometry | Fluorescence microscopy
Languages	English (Fluent) | German (Intermediate) | Tamil (Native) | Hindi (Beginner)

Academic Merits & Activities

Scholarships

2015	Birkigt Scholarship Award, ETH Zürich
2012	Undergraduate Research Scholarship, SASTRA University & Harvard – MIT HST

Mentoring

2016 – 2018	Tutored Bioprocess Modeling, Process Simulation and Flowsheeting and Statistical and Numerical Methods courses at ETH Zürich
2016 – 2024	Supervised doctoral and graduate students in research projects and semester theses at ETH Zürich, University at Buffalo and UZH Zürich